

Voice Classification Data

STAT380

2024-10-10

R Packages To Load

```
library(tuneR)
```

Set The Directory Where the Data Folder Is Located

```
dir_path="/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/"
```

Related Functions

Function for Extracting Features of a Single Audio File

```
Extract_Sound_Feature<-function(my_audio){  
  ss<-c( c(summary(my_audio@left)), SD=sd(my_audio@left))  
  names(ss)=c("Min", "Q1", "Median", "Mean", "Q3", "Max", "Sd")  
  return(ss)  
}
```

Function for Extracting Features from all Audio Files in a Folder

```
Extract_Sound_Feature_for_AllFile<-function(FolderName,  
                                             dir_path,  
                                             details=TRUE){  
  
  files_names<-list.files(paste0(dir_path,FolderName))  
  val_sound_feature=NULL  
  
  for(FileIndex in 1:length(files_names)){  
    file_path<-paste0(dir_path,FolderName,"/" , files_names[FileIndex])  
    sound_file<-readWave(file_path)  
    sound_feature<-Extract_Sound_Feature(sound_file)  
  
    if(!is.null(sound_feature)){val_sound_feature<-rbind(val_sound_feature, c(sound_fe  
    if(details){print(paste0("Sound File=",file_path, "' Loaded Succesfully. Feature I  
  }  
  
  Y=as.factor(rep(x = FolderName, dim(val_sound_feature)[1]))  
  return(data.frame(Y,val_sound_feature))  
}
```

Function To Create a Training Dataset

```
Create_Training_Data_Set<-function(Category1="females", Category2="males", dir_path, rand_err=20, details="none", browser=FALSE){
  #browser()
  Train_cate1<-Extract_Sound_Feature_for_AllFile(FolderName=Category1,dir_path=dir_path, details=details, browser=FALSE)
  Train_cate2<-Extract_Sound_Feature_for_AllFile(FolderName=Category2,dir_path=dir_path, details=details, browser=FALSE)
  data=rbind(Train_cate1, Train_cate2)
  #Y_var<-ifelse(data$Y==Category1, 1, 0)

  #vv<-round(runif(rand_err, 1,length(Y_var) ))
  #Y_var[vv]= 1- Y_var[vv]
  #data$Y= Y_var
  return(data)
}
```

Function for Extracting Features for a Testing Audio File

```
Extract_Feature_Test<-function(Test_File_Loc){
  file_path<-paste0(Test_File_Loc)
  sound_file<-readWave(file_path)
  sound_feature<-Extract_Sound_Feature(sound_file)
  return(as.data.frame(t(sound_feature)))
}
```

Fitting a Logistic Regression model and predict the Gender from the recorded audio file

```
library(tuneR)

Categories=c("males", "females")
Training_Sound_Data<-Create_Training_Data_Set(Category1 =Categories[1], Category2 = Categories[2], dir_path=dir_path, rand_err=20, details="none", browser=FALSE)

## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/0.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/1.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/10.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/11.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/12.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/13.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/14.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/15.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/16.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/17.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/18.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/19.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/2.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/20.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/21.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/22.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/23.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/24.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/25.wav' Loaded"
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/26.wav' Loaded"
```



```
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/30.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/31.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/32.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/33.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/34.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/35.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/36.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/37.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/38.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/39.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/4.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/40.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/41.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/42.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/43.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/44.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/45.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/46.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/47.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/48.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/49.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/5.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/50.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/51.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/6.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/7.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/8.wav' Load
## [1] "Sound File='/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/9.wav' Load

# fitting the model using a logistic regression model Female==1
fit<-glm(Y~., data=Training_Sound_Data, family="binomial")
```

Obtaining the predicted probability for a new file

Testing a Female Voice Sample

```
Test_File_Loc="/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/females/1.wav"

newData_feature=Extract_Feature_Test(Test_File_Loc)
predictedProb<-predict.glm(fit, newData=newData_feature, type="response")
paste0("The predicted probability that the selected file is:", "female", "( comparing between ", paste0(

## [1] "The predicted probability that the selected file is:female( comparing between males and females)

predictedProb_1=data.frame(1-predictedProb, predictedProb)
colnames(predictedProb_1)=Categories
rownames(predictedProb_1)="Predicted Group Probability"
predictedProb_1

##
## Predicted Group Probability 0.1678005 0.8321995
```

Testing a Male Voice Sample

```
Test_File_Loc="/Users/subhadippal/Desktop/STAT380/voice_modeling_starter-master/males/2.wav"

newData_feature=Extract_Feature_Test(Test_File_Loc)
predictedProb<-predict.glm(fit, newdata=newData_feature, type="response")
paste0("The predicted probability that the selected file is:", "female", "( comparing between ", paste0(

## [1] "The predicted probability that the selected file is:female( comparing between males and females.
predictedProb_1=data.frame(1-predictedProb, predictedProb)
colnames(predictedProb_1)=Categories
rownames(predictedProb_1)="Predicted Group Probability"
predictedProb_1

##
## Predicted Group Probability    males  females
```